



# UTM

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**BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)**

**SECR1213 NETWORK COMMUNICATIONS - SECTION 01**

**PROJECT TASK 3 - CORRECTION**

**CHOOSING THE APPROPRIATE LAN DEVICES**

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## 1.0 LAN Devices

### 1.1.1 Router

| Feature     | Cisco ISR 4331                                                                                                                                                                                                                                                                                                                                                      | TP-Link Archer AX6000                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance | 100Mbps and upgradeable to 300 Mbps                                                                                                                                                                                                                                                                                                                                 | Up to 6Gbps combined throughput (WiFi 6)                                                                                                                                                                                                                                                              |
| Modularity  | Highly modular (support WAN/LAN modules)                                                                                                                                                                                                                                                                                                                            | Fixed design, no modular options                                                                                                                                                                                                                                                                      |
| Security    | Enterprise-grade: Firewalls VPN, IPS                                                                                                                                                                                                                                                                                                                                | Consumer-grade: WPA3, antivirus, parental controls                                                                                                                                                                                                                                                    |
| Management  | Cisco DNA Center or CLI                                                                                                                                                                                                                                                                                                                                             | TP-Link Tether App                                                                                                                                                                                                                                                                                    |
| Scalability | Designed for large-scale networks                                                                                                                                                                                                                                                                                                                                   | Limited to small networks                                                                                                                                                                                                                                                                             |
| PoE Support | Yes                                                                                                                                                                                                                                                                                                                                                                 | No                                                                                                                                                                                                                                                                                                    |
| Price (RM)  | 4491.11                                                                                                                                                                                                                                                                                                                                                             | 1029.00                                                                                                                                                                                                                                                                                               |
| Pros        | <ul style="list-style-type: none"><li>- Enterprise-grade performance with modularity for scalability</li><li>- Advanced security features (VPN, firewall, threat detection)</li><li>- Reliable for handling complex enterprise networks</li><li>- Supports WAN optimization and SD-WAN</li><li>- Long-term support and enterprise-level service contracts</li></ul> | <ul style="list-style-type: none"><li>- Affordable, consumer-friendly pricing</li><li>- High-speed Wi-Fi 6 (up to 6 Gbps) for excellent wireless performance</li><li>- Easy to set up and manage with a mobile app</li><li>- Compact, sleek design for home or small office use</li></ul>             |
| Cons        | <ul style="list-style-type: none"><li>- Expensive, with high upfront costs</li><li>- Requires technical expertise to configure and manage</li><li>- No built-in wireless capability</li><li>- Bulky, rack-mount design</li><li>- Licensing costs for additional features</li></ul>                                                                                  | <ul style="list-style-type: none"><li>- Limited to fixed hardware, no expandability</li><li>- Basic security features compared to enterprise-grade routers</li><li>- Not suitable for large-scale enterprise networks</li><li>- No advanced routing capabilities for complex configurations</li></ul> |
| Resources   |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                   |

|  |                                                  |                                                                                                                                                           |
|--|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Cisco 4000 Family ISR Data Sheet</a> | <a href="https://www.tp-link.com/my/home-networking/wifi-router/archer-ax6000/">https://www.tp-link.com/my/home-networking/wifi-router/archer-ax6000/</a> |
|--|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

For routers we choose Cisco ISR 4331 instead of TP-Link Archer AX6000 because we need enterprise-grade performance, security, and scalability. Besides, Cisco can fulfill complex requirements or serve critical infrastructure. Next, we have a skilled IT team to configure and maintain the system.

### 1.1.2 Firewall

| Feature                           | Fortinet FortiGate 60F                                                                                                                                                                                                                                                                                                  | Cisco Firepower 1010                                                                                                                                                                                                                                                                        |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Firewall throughput               | 10Gbps                                                                                                                                                                                                                                                                                                                  | 1.9Gbps                                                                                                                                                                                                                                                                                     |
| Threat protection throughput      | 700Mbps                                                                                                                                                                                                                                                                                                                 | 650Mbps                                                                                                                                                                                                                                                                                     |
| Unified Threat Management UTM)    | Yes                                                                                                                                                                                                                                                                                                                     | Partial (requires additional licensing)                                                                                                                                                                                                                                                     |
| Ease of management                | Intuitive UI, FortiManager, FortiCloud                                                                                                                                                                                                                                                                                  | Requires expertise: FMC or CLI                                                                                                                                                                                                                                                              |
| Integration with existing network | Strong with SD-WAN and UTM focus                                                                                                                                                                                                                                                                                        | Seamless integration with Cisco products                                                                                                                                                                                                                                                    |
| Scalability                       | Excellent, can grow with campus needs                                                                                                                                                                                                                                                                                   | Scales well but higher costs for upgrades                                                                                                                                                                                                                                                   |
| Price (RM)                        | 4500.00                                                                                                                                                                                                                                                                                                                 | 2086.79                                                                                                                                                                                                                                                                                     |
| Pros                              | <ul style="list-style-type: none"> <li>- High performance with hardware-accelerated security features via SoC4 chip</li> <li>- Comprehensive threat protection (IPS, AV, web filtering)</li> <li>- Built-in SD-WAN for hybrid network environments</li> <li>- Simple subscription model for security updates</li> </ul> | <ul style="list-style-type: none"> <li>- Strong integration with Cisco networking infrastructure and tools</li> <li>- Advanced Threat Defense features (Malware Protection, URL Filtering, etc.)</li> <li>- Flexibility with support for multiple VPN types and advanced routing</li> </ul> |
| Cons                              | <ul style="list-style-type: none"> <li>- Limited to Fortinet ecosystem; integration with third-party solutions can be challenging</li> <li>- User interface may require learning for beginners</li> <li>- May not support as complex configurations as Cisco Firepower</li> </ul>                                       | <ul style="list-style-type: none"> <li>- Higher cost compared to similar SMB firewalls</li> <li>- Can be overkill for small, simple networks</li> <li>- Licensing model can be complex and expensive</li> <li>- Setup and management may require specialized Cisco knowledge</li> </ul>     |

|            |                                                                                                                                                |                                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| References | <br><a href="#">FortiGate FortiWiFi 60F Series Data Sheet</a> | <br><a href="#">Cisco Firepower 1010 NGFW Appliance</a> |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

For firewalls, we choose Fortinet FortiGate 60F instead of Cisco Firepower 1010 because Fortinet offers high performance at a more affordable price. Fortinet has user-friendly solutions for SMBs or enterprises without compromising on security. Fortinet also offers integrated UTM features and SD-WAN in a single device.

### 1.1.3 Wireless Access Point

|                 |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature         | Ubiquiti UniFi 6 Pro                                                                                                                                                                                                                                                                       | Cisco Catalyst 9105AXI                                                                                                                                                                                                                                                           |
| Wi-Fi Standard  | Wi-Fi 6 (802.11ax)                                                                                                                                                                                                                                                                         | Wi-Fi 6 (802.11ax)                                                                                                                                                                                                                                                               |
| Max Throughput  | Up to 4.8 Gbps                                                                                                                                                                                                                                                                             | Up to 5.38 Gbps                                                                                                                                                                                                                                                                  |
| Frequency Bands | Dual-band (2.4 GHz and 5 GHz)                                                                                                                                                                                                                                                              | Tri-band (2.4 GHz, 5 GHz, and additional 5 GHz)                                                                                                                                                                                                                                  |
| PoE Support     | PoE and PoE+                                                                                                                                                                                                                                                                               | PoE+                                                                                                                                                                                                                                                                             |
| Security        | WPA3, VLAN support, and guest isolation                                                                                                                                                                                                                                                    | WPA3, advanced identity-based networking, and encryption                                                                                                                                                                                                                         |
| Scalability     | Easy to scale for medium networks                                                                                                                                                                                                                                                          | Enterprise-grade scalability for larger environments                                                                                                                                                                                                                             |
| Price           | 1024.00                                                                                                                                                                                                                                                                                    | 1900.08                                                                                                                                                                                                                                                                          |
| Pros            | <ul style="list-style-type: none"> <li>- Affordable, excellent value for small to medium-sized networks</li> <li>- Wi-Fi 6 support for higher speeds and better performance in dense environments</li> <li>- Easy to set up and manage with UniFi Controller or UniFi Cloud Key</li> </ul> | <ul style="list-style-type: none"> <li>- Enterprise-grade performance with Wi-Fi 6 capabilities</li> <li>- Advanced security features (Secure Boot, Trustworthy Solutions, rogue AP detection)</li> <li>- Multi-gigabit Ethernet support for higher-speed connections</li> </ul> |
| Cons            | <ul style="list-style-type: none"> <li>- Limited advanced enterprise features compared to Cisco</li> <li>- Lacks built-in advanced security features</li> <li>- No multi-gigabit Ethernet port for higher throughput in future-proofing</li> </ul>                                         | <ul style="list-style-type: none"> <li>- Expensive, suitable for enterprise budgets</li> <li>- Requires Cisco expertise for configuration and management</li> <li>- Overkill for small to medium-sized networks without enterprise-level requirements</li> </ul>                 |
| Reference       | <br><a href="#">Ubiquiti UniFi 6 Pro</a>                                                                                                                                                                | <br><a href="#">Cisco catalyst 9105AX</a>                                                                                                                                                    |

For wireless access point, we choose Ubiquiti Unifi 6 Pro because it also suitable for labs, classrooms, and open study spaces where moderate client density (20–50 devices per AP) is expected.

### 1.1.4 Switches (48 ports)

|                      |                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                            |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature              | Cisco Catalyst 2960-X (48-Port)                                                                                                                                                                                                                                                     | Ubiquiti UniFi Switch Pro 48 (48-Port)                                                                                                                                                                                                                                     |
| Ports                | 48 Gigabit Ethernet (GE) + 4 SFP uplinks (Gigabit)                                                                                                                                                                                                                                  | 48 Gigabit Ethernet (GE) + 2 10G SFP+ uplinks                                                                                                                                                                                                                              |
| PoE Support          | PoE+ (up to 740W)                                                                                                                                                                                                                                                                   | PoE+ (up to 400W)                                                                                                                                                                                                                                                          |
| Layer 3 Capabilities | Basic Layer 3 (Static routing)                                                                                                                                                                                                                                                      | Layer 3 support via advanced software with UniFi Controller                                                                                                                                                                                                                |
| Switching Capacity   | 216 Gbps (Non-blocking)                                                                                                                                                                                                                                                             | 70 Gbps (Non-blocking)                                                                                                                                                                                                                                                     |
| Performance          | High-performance with industry-leading switching capacity                                                                                                                                                                                                                           | Solid performance for small to medium-sized environments                                                                                                                                                                                                                   |
| Price                | 700.00                                                                                                                                                                                                                                                                              | 2700.00                                                                                                                                                                                                                                                                    |
| Pros                 | <ul style="list-style-type: none"> <li>- Advanced security features (802.1x, MACsec, TrustSec support)</li> <li>- Supports Energy Efficient Ethernet (EEE) for lower power consumption</li> <li>- Robust support for Layer 2 features with some Layer 3 (static routing)</li> </ul> | <ul style="list-style-type: none"> <li>- Supports Layer 3 capabilities for inter-VLAN routing and static routing</li> <li>- Offers both fiber (SFP+) and copper connectivity options</li> <li>- Sleek, modern design and compatibility with the UniFi ecosystem</li> </ul> |
| Cons                 | <ul style="list-style-type: none"> <li>- Complex setup and management may require Cisco expertise</li> <li>- Lacks modern features like centralized cloud management</li> <li>- Limited to Cisco's proprietary ecosystem for optimal use</li> </ul>                                 | <ul style="list-style-type: none"> <li>- Less robust warranty and support compared to Cisco</li> <li>- Not as feature-rich for Layer 3 and advanced routing needs</li> <li>- Limited advanced enterprise features compared to Cisco</li> </ul>                             |
| Reference            | <br><a href="#">Cisco Catalyst 2960-X</a>                                                                                                                                                        | <br><a href="#">Ubiquiti Networks UniFi Switch PRO 48</a>                                                                                                                              |

For 48 ports switches, we choose Cisco Catalyst 2960-X instead of Ubiquiti UniFi Switch Pro 48 because it is affordable and offers scalability for a faculty setup. The high PoE (740W) is sufficient for powering devices like access points or IP cameras. It can integrate well into enterprise networks as our requirement about the network is mission-critical since it requires cutting-edge technology, or integrates heavily into a Cisco ecosystem.



### 1.1.5 Switches (24 ports)

|                      |                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature              | Cisco Catalyst 2960-L (24-Port)                                                                                                                                                                                                                       | Ubiquiti UniFi Switch 24 (24-Port)                                                                                                                                                                                                          |
| Ports                | 24 Gigabit Ethernet (GE) ports + 4 Gigabit SFP uplinks                                                                                                                                                                                                | 24 Gigabit Ethernet (GE) ports + 2 Gigabit SFP uplinks                                                                                                                                                                                      |
| PoE Support          | PoE+ (up to 370W)                                                                                                                                                                                                                                     | PoE+ (up to 400W)                                                                                                                                                                                                                           |
| Layer 3 Capabilities | Layer 2 switching (Basic Layer 3 for static routing)                                                                                                                                                                                                  | Layer 2 switching (Layer 3 features via UniFi Controller)                                                                                                                                                                                   |
| Switching Capacity   | 56 Gbps (Non-blocking)                                                                                                                                                                                                                                | 52 Gbps (Non-blocking)                                                                                                                                                                                                                      |
| Price                | 700.00                                                                                                                                                                                                                                                | 1776.30                                                                                                                                                                                                                                     |
| Pros                 | <ul style="list-style-type: none"> <li>- Advanced security features (802.1x, MAC filtering, etc.)</li> <li>- Energy-efficient design (EEE) to reduce power consumption</li> <li>- Built for long-term stability in enterprise environments</li> </ul> | <ul style="list-style-type: none"> <li>- Centralized cloud management through UniFi Controller</li> <li>- User-friendly interface suitable for SMBs and home office setups</li> <li>- Modern design with options for SFP uplinks</li> </ul> |
| Cons                 | <ul style="list-style-type: none"> <li>- Lacks advanced Layer 3 capabilities (primarily Layer 2 focused)</li> <li>- Proprietary ecosystem with limited third-party integrations</li> <li>- Setup and management require Cisco expertise</li> </ul>    | <ul style="list-style-type: none"> <li>- Relies heavily on UniFi ecosystem for seamless integration</li> <li>- Less suited for highly secure or enterprise-critical deployments</li> <li>- Less robust support compared to Cisco</li> </ul> |
| Reference            | <br><a href="#">Cisco Catalyst 2960-L Series Switches Data Sheet - Cisco</a>                                                                                       | <br><a href="#">Switch 24 PoE (250W) - Ubiquiti Store United States</a>                                                                                 |

For 24 ports switches, we choose Cisco Catalyst 2960-L instead of Ubiquiti UniFi Switch 24 because they offer great value for money, strong security, and are suitable for larger-scale deployments like classrooms, labs, and administrative offices. These factors, combined with their affordability and long-term reliability, make them a perfect fit for our needs.

### 1.1.6 Patch Panel (48 ports)

|                     |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                  |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Features            | Tripp Lite N252-048-1U (48 ports)                                                                                                                                                                                                 | TRENDnet TC-P48C6 (48 ports)                                                                                                                                                                                     |
| Ports               | 48 RJ45 Cat5e/6 ports                                                                                                                                                                                                             | 48 RJ45 Cat6 ports                                                                                                                                                                                               |
| Cable Compatibility | Compatible with Cat5e and Cat6 cables                                                                                                                                                                                             | Compatible with Cat6 cables, but works with lower categories too                                                                                                                                                 |
| Material            | Heavy-duty steel                                                                                                                                                                                                                  | Steel                                                                                                                                                                                                            |
| Price               | 1191.70                                                                                                                                                                                                                           | 311.30                                                                                                                                                                                                           |
| Pros                | <ul style="list-style-type: none"> <li>- Ideal for network setups requiring reliable performance and organization</li> <li>- Designed for easy cable management and organization with labeling options</li> </ul>                 | <ul style="list-style-type: none"> <li>- Simple installation and well-labeled ports for easier cable organization</li> <li>- Compact design, fits easily into 19-inch racks</li> </ul>                           |
| Cons                | <ul style="list-style-type: none"> <li>- Lacks some advanced features like modularity or higher-density designs</li> <li>- May not be as durable or flexible as premium brands like Leviton for demanding environments</li> </ul> | <ul style="list-style-type: none"> <li>- Limited advanced features for future-proofing or scalability</li> <li>- Some users report cable management can be slightly more difficult compared to others</li> </ul> |
| Reference           | <br><a href="#">Tripp Lite Patch Panel 48 port</a>                                                                                             | <br><a href="#">48-port Cat6 Unshielded Patch Panel - TRENDnet TC-P48C6</a>                                                  |

For 48 ports patch panel, we choose Tripp Lite N252-048-1U instead of TRENDnet TC-P48C6 because of its high-density design, enterprise-grade durability, and integrated cable management, ideal for large-scale deployments.

### 1.1.7 Patch Panel (24 ports)

|                     |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Features            | Tripp Lite N252-024 (24-Port)                                                                                                                                                                                                   | Legrand 24-Port Cat6 Patch Panel (24 ports)                                                                                                                                                                                            |
| Ports               | 24 Cat6 RJ45 ports                                                                                                                                                                                                              | 24 Cat6 RJ45 ports                                                                                                                                                                                                                     |
| Cable Compatibility | Compatible with Cat5e and Cat6 cables                                                                                                                                                                                           | Compatible with Cat5e, Cat6, and lower category cables                                                                                                                                                                                 |
| Material            | Heavy-duty metal construction                                                                                                                                                                                                   | Metal construction                                                                                                                                                                                                                     |
| Price               | 241.20                                                                                                                                                                                                                          | 800.55                                                                                                                                                                                                                                 |
| Pros                | <ul style="list-style-type: none"> <li>- Sturdy construction with a reasonable level of durability</li> <li>- Affordable and good value for small to medium-sized network installations</li> </ul>                              | <ul style="list-style-type: none"> <li>- Excellent build quality with robust materials, ensuring durability and longevity</li> <li>- Comes with solid cable management options, reducing clutter and improving organization</li> </ul> |
| Cons                | <ul style="list-style-type: none"> <li>- Limited advanced features such as enhanced cable management tools or modularity</li> <li>- May not be suitable for high-density or high-performance enterprise environments</li> </ul> | <ul style="list-style-type: none"> <li>- Installation might require more time due to better cable management features</li> <li>- May not offer additional ports or expandability options for larger, more complex networks</li> </ul>  |
| Reference           | <br><a href="#">Tripp Lite 24-Port 1U Rackmount Cat6 110 Patch Panel 568B, RJ45 Ethernet(N252-024)</a>                                       | <br><a href="#">Cat 6 24-Port Rack Mount Patch Panel</a>                                                                                           |

For 24 ports patch panel, we choose Tripp Lite N252-024 instead of Legrand 24-Port Cat6 Patch Panel because it is suitable for basic deployments with moderate cable management. It is ideal for small to medium-sized network setups

### 1.1.8 Server

| Feature         | Cisco UCS C480 M5                                                                                                                                                                                                                                                                                    | Lenovo ThinkSystem SR630                                                                                                                                                                                                                     |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processor       | Up to 4 Intel Xeon Scalable processors                                                                                                                                                                                                                                                               | Up to 2 Intel Xeon Scalable processors                                                                                                                                                                                                       |
| Memory          | Up to 6 TB DDR4 memory, 24 DIMM slots                                                                                                                                                                                                                                                                | Up to 1.5 TB DDR4 memory, 16 DIMM slots                                                                                                                                                                                                      |
| Storage Options | Up to 12x 3.5" or 2.5" drives with flexible configurations                                                                                                                                                                                                                                           | Up to 10x 2.5" or 12x 3.5" drives                                                                                                                                                                                                            |
| Expansion Slots | 8 PCIe 3.0 slots                                                                                                                                                                                                                                                                                     | 6 PCIe 3.0 slots                                                                                                                                                                                                                             |
| Power Supply    | 2x 1600W or 2x 2000W redundant power supplies                                                                                                                                                                                                                                                        | 2x 550W, 750W or 1100W redundant power supplies                                                                                                                                                                                              |
| Price           | 39441.75                                                                                                                                                                                                                                                                                             | 15859.00                                                                                                                                                                                                                                     |
| Pros            | <ul style="list-style-type: none"> <li>- Integrated management with Cisco Intersight, enabling cloud-based monitoring and management</li> <li>- Excellent for mission-critical applications and large data centers</li> <li>- Supports a wide range of storage options (local, SAN, etc.)</li> </ul> | <ul style="list-style-type: none"> <li>- Excellent value for performance, scalability, and price, suitable for a wide range of workloads</li> <li>- Simplified management with Lenovo XClarity for easy deployment and monitoring</li> </ul> |
| Cons            | <ul style="list-style-type: none"> <li>- Limited flexibility in terms of supporting other ecosystems outside of Cisco's proprietary tech</li> <li>- More complex to deploy and manage without prior experience with Cisco UCS</li> </ul>                                                             | <ul style="list-style-type: none"> <li>- Fewer advanced integration features compared to Cisco, particularly for larger environments</li> <li>- May require additional purchases for advanced management and monitoring options</li> </ul>   |
| Reference       | <br><a href="#">Cisco UCS C480 M5 High-Performance Rack Server</a>                                                                                                                                                | <br><a href="#">Lenovo ThinkSystem SR630</a>                                                                                                             |

For server, we choose Cisco UCS C480M5 instead of Lenovo ThinkSystem SR630 because Cisco UCS C480M5 is a more powerful and scalable server. It boasts a higher maximum CPU capacity, enabling it to handle more demanding workloads, and a higher maximum memory

capacity, which can help to improve performance for memory-intensive applications. It also offers more storage options, providing the flexibility to store more data and choose the most suitable storage type.

### 1.1.9 Connector

|                      |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature              | Dintek Cat 6 Unshielded RJ45 Connector                                                                                                                                                                                                                  | AMP Tyco RJ45 CAT6 8P8C Unshielded Modular Plug Network Lan Connector                                                                                                                                                                                                                                                          |
| Ease of Installation | Straightforward; basic design for standard crimping.                                                                                                                                                                                                    | Includes user-friendly features like guides and alignment tools for easier installation.                                                                                                                                                                                                                                       |
| Performance          | Meets Cat 6 standards; suitable for up to 10 Gbps over short distances.                                                                                                                                                                                 | Meets or exceeds Cat 6 standards; offers better long-term reliability and consistency.                                                                                                                                                                                                                                         |
| Build quality        | Good for typical installations but less robust in high-stress environments.                                                                                                                                                                             | Superior build quality; designed for mission-critical and high-performance applications.                                                                                                                                                                                                                                       |
| Application          | Small to medium-scale networks; cost-sensitive projects.                                                                                                                                                                                                | Enterprise networks, data centers, or environments requiring high durability and precision.                                                                                                                                                                                                                                    |
| Price (RM)           | 3.00                                                                                                                                                                                                                                                    | 24.90 (100pcs)                                                                                                                                                                                                                                                                                                                 |
| Pros                 | <ul style="list-style-type: none"> <li>- Affordable and widely available for various networking projects</li> <li>- Easy to use with standard crimping tools</li> <li>- Available in bulk, making it cost-effective for larger installations</li> </ul> | <ul style="list-style-type: none"> <li>- High-quality build and reliability, ideal for professional and enterprise-level installations</li> <li>- Designed for easy crimping with robust connectors to ensure secure connections</li> <li>- Offers consistent and dependable performance for network infrastructure</li> </ul> |
| Cons                 | <ul style="list-style-type: none"> <li>- May not be as durable or reliable in high-performance or critical network environments</li> <li>- May not provide the best performance in high-density or interference-heavy</li> </ul>                        | <ul style="list-style-type: none"> <li>- Not as widely available as more common RJ45 connectors</li> <li>- Limited options compared to other brands offering similar products at lower prices</li> </ul>                                                                                                                       |

|           |                                                                                                                                                                                                                             |                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | environments                                                                                                                                                                                                                |                                                                                                                                                                             |
| Reference | <br><a href="#">Dintek Cat.6 Unshielded RJ45 Modular Plug, 150188027   Asashi Technology Sdn Bhd (332541-T)   Malaysia IT Online Store</a> | <br><a href="#">AMP Tyco RJ45 CAT6 8P8C Modular Plug Network Lan Connector (100 PCS)</a> |

For LAN connectors, we choose the AMP Tyco RJ45 CAT6 8P8C Modular Plug Network LAN Connector instead of the Dintek Cat 6 Unshielded RJ45 Connector because AMP Tyco offers superior build quality and long-term reliability. Its user-friendly features, such as guides and alignment tools, make installation more precise and efficient. Additionally, AMP Tyco exceeds Cat 6 standards, ensuring consistent performance and making it suitable for high-performance applications like enterprise networks and data centers. The 100-piece bundle also provides cost-effectiveness for large-scale implementations.

### 1.1.10 Cable

|              |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature      | trueCable<br>6ECMRBLU_1Kbx                                                                                                                                                                                                                                                    | Cat 6 Ethernet Cable 305m<br>with 23 AWG and 4 twisted<br>pairs                                                                                                                                                                                                                                                     |
| Bandwidth    | Up to 550 MHz, ensuring<br>enhanced data transmission.                                                                                                                                                                                                                        | Up to 550 MHz, similar<br>high-frequency support.                                                                                                                                                                                                                                                                   |
| Speed        | Supports 10 Gbps up to 165<br>feet (50m); drops after                                                                                                                                                                                                                         | Supports 10 Gbps over short<br>distances.                                                                                                                                                                                                                                                                           |
| Material     | Solid bare copper, more<br>durable but costlier.                                                                                                                                                                                                                              | Copper-Clad Aluminum<br>(CCA), cost-effective.                                                                                                                                                                                                                                                                      |
| Jacket Type  | CMR (Riser-rated), ideal for<br>fire safety in vertical<br>installations.                                                                                                                                                                                                     | Standard, not specialized for<br>fire-resistance.                                                                                                                                                                                                                                                                   |
| Ease of Use  | Pull-box packaging, similar<br>ease of handling.                                                                                                                                                                                                                              | Standard pull-box<br>packaging for installation.                                                                                                                                                                                                                                                                    |
| PoE Support  | Supports PoE++ (up to<br>100W), better for advanced<br>setups                                                                                                                                                                                                                 | Compatible with PoE and<br>PoE+ devices.                                                                                                                                                                                                                                                                            |
| Applications | Ideal for professional and<br>enterprise environments.                                                                                                                                                                                                                        | Suitable for general-purpose<br>networks.                                                                                                                                                                                                                                                                           |
| Price (RM)   | 783.50                                                                                                                                                                                                                                                                        | 296.50                                                                                                                                                                                                                                                                                                              |
| Pros         | <ul style="list-style-type: none"> <li>- High-quality construction, ideal for professional and commercial use</li> <li>- Solid build with extra protection against noise and interference</li> <li>- UL-listed and RoHS compliant, ensuring safety and reliability</li> </ul> | <ul style="list-style-type: none"> <li>- Affordable, providing good value for large installations and high-performance networks</li> <li>- Available in bulk (305 meters) for extensive installations</li> <li>- 23 AWG wires offer better signal transmission over longer distances than thinner cables</li> </ul> |
| Cons         | - Bulk options may not be as                                                                                                                                                                                                                                                  | - May not provide the same                                                                                                                                                                                                                                                                                          |



|           |                                                                                                                                   |                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | versatile for all networking environments<br>- Can be more difficult to work with compared to lower-gauge cables due to stiffness | longevity and robustness for industrial or mission-critical applications<br>- Less optimized for environments with heavy electromagnetic interference (EMI) |
| Reference | <br><a href="#">true cable</a>                   | <br><a href="#">CAT 6 Ethernet Cable 305m</a>                            |

The Cat 6 Ethernet Cable 305m is a better choice for its affordability and sufficient performance, offering 550 MHz bandwidth and 10 Gbps speed. It supports PoE devices and is ideal for small to medium networks. While trueCABLE provides a fire-resistant jacket and solid copper, these features add unnecessary cost for general setups, making the Cat 6 Ethernet Cable 305m more practical.

## 1.2 Expected cost

| No.             | Device                                                                        | Quantity | Price per unit (RM) | Total Unit Price (RM) |
|-----------------|-------------------------------------------------------------------------------|----------|---------------------|-----------------------|
| 1.              | Cisco ISR 4331 routers                                                        | 1        | 4491.11             | 4491.11               |
| 2.              | Fortinet FortiGate 60 F firewalls                                             | 1        | 4500.00             | 4500.00               |
| 3.              | Ubiquiti Unifi 6 Pro wireless access point                                    | 4        | 1024.00             | 4096.00               |
| 4.              | Cisco Catalyst 2960-X for 48 ports switches                                   | 2        | 700.00              | 1400.00               |
| 5.              | Cisco Catalyst 2960-L for 24 ports switches                                   | 1        | 700.00              | 700.00                |
| 6.              | Tripp Lite N252-048-1U for 48 ports patch panel                               | 2        | 1191.70             | 2383.4                |
| 7.              | Tripp Lite N252-024 for 24 ports patch panel                                  | 2        | 241.20              | 482.40                |
| 8.              | Cisco UCS C480M5 server                                                       | 5        | 39441.75            | 197208.75             |
| 9.              | AMP Tyco RJ45 CAT6 8P8C Unshielded Modular Plug Network Lan Connector(100pcs) | 9        | 24.90               | 224.10                |
| 10.             | Cat 6 Ethernet Cable 305m with 23 AWG and 4 twisted pairs                     | 9        | 296.50              | 2668.50               |
| Total cost (RM) |                                                                               |          |                     | 218154.26             |

## 2.0 Report - Questions and Answers

### 1. Are you surprised by the prices? How were you surprised?

Answer: Yes, the prices of networking devices can be quite surprising. As we don't know that the devices can be quite expensive before we do research for this project, especially when exploring enterprise-grade equipment like Cisco routers and switches. The high cost of these devices often reflects their advanced features, reliability, and brand reputation. Another surprising aspect is the variance in pricing between brands and models. While premium brands like Cisco offer cutting-edge technology and top-tier support, budget-friendly options still meet the needs of less demanding environments at a fraction of the cost. The realization that these prices also account for factors like build quality, long-term durability, and enhanced security features can help make sense of the numbers, even though they still seem steep at first glance.

### 2. Have you ever considered cost as a factor for choosing networking devices?

Answer: For us, cost is always an important factor when selecting networking devices, but it's not the only one. As our group's budget is RM 2 millions, hence balancing the budget with the specific needs of the network is critical. For instance, while it's tempting to go for cheaper devices to save costs, it's important to think about the long-term implications, such as durability, scalability, and maintenance expenses. Total Cost of Ownership (TCO) is a key consideration—cheaper devices may save money upfront, but premium devices with better warranties and support often prove more economical in the long run. In an academic setup, prioritizing cost-effective devices for general areas while investing in high-performance ones for specialized labs can strike the right balance. For example, using TP-Link or Ubiquiti for less demanding environments and reserving Cisco for high-tech labs allows us to manage costs effectively while still ensuring quality.

### 3. What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Answer:

#### **Router: Cisco vs Tp-Link**

For target audience, Cisco is designed for enterprise-level networks, businesses, and critical infrastructure while Tp-Link is primarily aimed at home users, small businesses, and budget-conscious buyers.

For performance, Cisco has high performance as it is built for high-throughput and low-latency operations in demanding environments meanwhile Tp-Link has good performance as it is adequate for home and small office environments but may struggle under heavy loads..

For security, Cisco provides enterprise-grade security features such as intrusion prevention, advanced firewalls, and zero-trust architecture while Tp-Link provides basic security features such as WPA3, firewalls, and basic VPNs for consumer and small business use.

For price, Cisco tends to be more expensive to reflect its focus on enterprise and critical systems meanwhile Tp-Link is more affordable as it focuses on delivering value for money for home and small business users.

### **Switch: Cisco vs Huawei**

The popular series of Cisco switches are the enterprise switches, Catalyst series and Nexus switches. Cisco Catalyst switches are designed for core layers in campus networks, while the Nexus is mainly for data centers. Whereas, for Huawei switches, the popular one is the campus switches.

For target audience, Cisco is geared toward enterprises, data centers, service providers, and mission-critical networks meanwhile Huawei aims to serve enterprises, SMBs, and telecom operators with cost-effective solutions.

For performance, both Cisco and Huawei brands provide high-performance switches. Cisco uses IOS for stability and advanced management, while Huawei employs VRP which offers flexibility and traffic optimization.

For cost, Cisco switches are more expensive which reflects their premium features and reliability. Whereas, Huawei provides cost-effective solutions, appealing to budget-conscious users.

### **Patch panel: Tripp Lite vs TRENDnet**

For target audience, Tripp Lite is designed for enterprise-grade and data center environments while TRENDnet is aimed at small to medium-sized businesses (SMBs) and budget-conscious users.

For its features, Tripp Lite offers high-density patch panels, supporting large-scale networks meanwhile TRENDnet provides standard patch panels, often for Cat5e and Cat6 cables and focused on basic features like numbered ports and simple design.

For shielding and performance, TRENDnet emphasizes shielding for environments with potential EMI, whereas Tripp Lite patch panels are more general-purpose, catering to standard networking needs.

For PoE compatibility, Tripp Lite frequently includes PoE features which may not be standard in all TRENDnet models.

For build and design, TRENDnet often uses metal shielding for added durability and stability, while Tripp Lite focuses on flexible setups with easy cable management.

For price, Tripp Lite offers a higher price point, reflecting its professional-grade quality and features, while TRENDnet is more affordable and budget-friendly.

### **Server : Cisco UCS C480M5 High-Performance Rack Server VS Lenovo ThinkSystem SR630**

The Cisco UCS C480M5 High-Performance Rack Server is a 4U rackmount system powered by dual 2nd-generation Intel Xeon Scalable processors meanwhile Lenovo ThinkSystem SR630 is a 2U rackmount server designed with up to 2nd Gen Intel Xeon Scalable processors, supporting configurations with up to 28 cores.

For memory capacity, Cisco UCS C480M5 supports up to 24 DDR4 DIMM slots, allowing a maximum memory capacity of 384GB meanwhile Lenovo ThinkSystem SR630 provides up to 12 DDR4 DIMM slots, enabling a maximum memory capacity of 192GB.

For storage, Cisco UCS C480M5 accommodates up to 32 2.5-inch hot-swappable SAS/SATA HDDs or SSDs, and it includes dual 10GBASE-T Intel x550 network

adapters meanwhile Lenovo ThinkSystem SR630 allows up to 16 2.5-inch hot-swappable SAS/SATA HDDs or SSDs, and its networking capabilities include dual 10GbE SFP+ ports.

For its additional feature, Cisco UCS C480M5 also supports up to 12 NVMe SSDs and features an integrated management controller (iMC) meanwhile Lenovo ThinkSystem SR630 supports up to 6 NVMe SSDs and offers management through the BMC with IPMItool.

### Meeting Minutes #6

|               |                                                             |                                                                                                                                                   |                    |
|---------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| DATE/TIME     | 2 Dec 2024 10:00 am                                         |                                                                                                                                                   |                    |
| LOCATION      | Physical                                                    |                                                                                                                                                   |                    |
| AGENDA        | 1. Discuss the question on Task 3                           |                                                                                                                                                   |                    |
|               | 2. Task distribution for each person                        |                                                                                                                                                   |                    |
|               | 3. Research on network devices for the academic institution |                                                                                                                                                   |                    |
| MEETING MC    | KOO XUAN                                                    |                                                                                                                                                   |                    |
| ATTENDANCE    |                                                             |                                                                                                                                                   |                    |
| NAME          | TIME                                                        |                                                                                                                                                   | REASON FOR ABSENCE |
| CHIN PEI WEN  | 1000                                                        |                                                                                                                                                   | -                  |
| KOO XUAN      | 1000                                                        |                                                                                                                                                   | -                  |
| LING YU QIAN  | 1000                                                        |                                                                                                                                                   | -                  |
| TAN ZHAO HONG | 1000                                                        |                                                                                                                                                   | -                  |
| Minutes       |                                                             |                                                                                                                                                   |                    |
| No.           | Item Discussed                                              | Details                                                                                                                                           | Person-In-Charge   |
| 1             | Understand the task requirement                             | Slide showed the question and rubric given on Task 3                                                                                              | All members        |
| 2             | Research on network devices for the academic institution    | All members discussed the importance of selecting appropriate network devices for the academic institution.                                       | All members        |
|               |                                                             | Members shared their research findings on routers, switches, wireless devices, and patch panels from various brands like Cisco, Huawei, and Asus. |                    |

|   |                                                           |                                                                                                                                         |             |
|---|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3 | Selection of LAN Devices Based on Requirements and Budget | We state out the specifications and features to identify devices for our project.                                                       | All members |
|   |                                                           | Finding the prices of the devices we selected                                                                                           |             |
|   |                                                           | We choose and select the most suitable devices for the project.                                                                         |             |
|   |                                                           | All members agreed to the suggested devices based on our budget.                                                                        |             |
|   |                                                           | Arrange the references used by all members.                                                                                             |             |
| 4 | Device Comparison Between Different Brands                | Members discussed the major differences between different brands of devices which focused on their performance, scalability, and price. | All members |
| 5 | Meeting ended                                             | 1230                                                                                                                                    | All members |

### Meeting Minutes #7

|               |                                    |                                                           |                    |
|---------------|------------------------------------|-----------------------------------------------------------|--------------------|
| DATE/TIME     |                                    | 16 Dec 2024 1:00 pm                                       |                    |
| LOCATION      |                                    | Physical                                                  |                    |
| AGENDA        |                                    | Correction - Task 3                                       |                    |
| MEETING MC    |                                    | LING YU QIAN                                              |                    |
| ATTENDANCE    |                                    |                                                           |                    |
| NAME          |                                    | TIME                                                      | REASON FOR ABSENCE |
| CHIN PEI WEN  |                                    | 1300                                                      | -                  |
| KOO XUAN      |                                    | 1300                                                      | -                  |
| LING YU QIAN  |                                    | 1300                                                      | -                  |
| TAN ZHAO HONG |                                    | 1300                                                      | -                  |
| Minutes       |                                    |                                                           |                    |
| No.           | Item Discussed                     | Details                                                   | Person-In-Charge   |
| 1             | Identify mistakes for each part    | We refer to the comments that were given by the lecturer. | All members        |
| 2             | Suggestion to correct the mistake. | All members discussed to add more references and citation | All members        |
| 5             | Meeting ended                      | 1430                                                      | All members        |



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